When the biomarker is also prognostic: bias in the biomarker-treatment interaction test of aggregated N-of-1 trials and two remedies

First draft, assembled from the material that left papers 01 and 02

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Abstract

Background. The biomarker-treatment interaction test of an aggregated N-of-1 trial is built for a predictive biomarker. A biomarker can also be prognostic. It can then be associated with the natural history, the placebo response or the baseline level. We asked when such an association biases the interaction test, and what removes the bias.

Methods. We simulated the Hybrid design under the extended data-generating process (DGP) of the companion DGP paper (paper 01), with the parameter values published by Hendrickson et al. Five prognostic pathways were examined. Three are correlations of the biomarker with a component. Two are mean shifts of the natural-history component, one constant and one growing over the trial. We compared analyses with and without the design-expectancy terms η_t and $B_i^c \eta_t$. A second stress test, run with the driver of paper 01, fitted ten analyses to trials with four prognostic pathways at strength 0.3. Its analyses keep the baseline visit as an outcome row.

Results. Under a correlation with the placebo-response component of 0.45, the analyses without expectancy terms rejected a true null at 0.092 (G9) at $N = 280$. With the terms, the rate was 0.060 (F1). The two-MCSE band at 1,000 replicates was $[0.036, 0.064]$. Under a natural-history shift that grows over the trial, G9 rejected at 0.073 and F1 at 0.052 at $N = 70$,

with no baseline row in the analysis. Without a prognostic effect, F1A0 reached 98.2 percent of the power of G10 in the Hybrid cells. In the second stress test, the analyses with expectancy terms stayed inside the two-MCSE band under every pathway. Every analysis without them, G10 included, exceeded the band under at least one pathway. The bias from the growing shift changed sign when a baseline row was added. The power loss once attributed to a strong prognostic correlation came from repaired covariance matrices. Under the current stop on infeasible matrices, all 10 of those cells were refused.

Conclusions. For a biomarker that may be prognostic, add η_t and $B_i^c \eta_t$ to the recommended analysis of the companion carryover paper (paper 02). A biomarker that may predict the rate of natural-history change needs a biomarker-by-time term. That remedy has not yet been simulated.

Notation and glossary

Symbols follow `analysis/report/NOTATION.md`. Two symbols are not yet listed there and are marked *proposed*.

Symbol	Meaning
B_i, B_i^c	biomarker of participant i; centered at the trial's baseline mean
z_i	standardized biomarker
w_t	calendar week of visit t
c_{bm}	on-drug biomarker-response correlation, the interaction under the correlation encoding
$c_{bm,PB}$, $c_{bm,TV}$, $c_{bm,BL}$	prognostic correlations of the biomarker with the placebo-response component, the natural-history component and the baseline level
b_{TV} (*proposed*)	constant natural-history mean shift per SD of the biomarker
b_{TV}^{tr} (*proposed*)	coefficient of the trend pathway: a shift of $b_{TV}^{\mathrm{tr}}\,\sigma_{TV}\,z_i\,g(w_t)$, with g the natural-history mean course scaled to peak 1
σ_{BR}, σ_{PB}, σ_{TV}	SDs of the drug-response, placebo-response and natural-history components
$t_{1/2}$, $t_{1/2}^a$	carryover half-life of the DGP; half-life assumed by the analysis
$D_{bc,it}$	exposure-weighted drug indicator built at $t_{1/2}^a$
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline
$\beta_{bm:D}$	interaction coefficient, the target of the test

Symbol	Meaning
β_η, $\beta_{bm:\eta}$	coefficients of η_t and $B_i^c\eta_t$
κ	mean model standard error divided by the empirical SD of the estimates
$\hat{\pi}$, MCSE	estimated rejection rate and its Monte Carlo standard error

- **Correlation encoding.** The interaction is a correlation between the biomarker and the drug response. It is present on drug and decays after discontinuation at the carryover half-life. Code labels it `mvn`.
- **Original DGP; extended DGP.** The original DGP is the code published by Hendrickson et al. at commit `58b32a9` (1). The extended DGP is the one recommended in paper 01 (2), which calls it the recommended DGP. It has a decaying biomarker-response correlation, the same first-order autoregressive correlation for every component with a constant between-component correlation at every lag ($\rho = 0.7$, $c_1 = 0.2$), carryover that decays from the last on-drug mean, and a stop on correlation matrices that are not positive definite. Code and data label it `refined`.
- **Published parameter set; simcore2026 set.** The published set is the component means and SDs of Hendrickson et al. Paper 01 calls them the default values. The simcore2026 set is an earlier set used by the paper 02 drivers before 2026-11-03. Runs under it are labeled historical in this draft.
- **Predictive biomarker.** One that changes the drug response ($c_{bm} \neq 0$). **Prognostic biomarker:** one associated with the outcome through another channel. A biomarker can be both.
- **Prognostic pathway.** One channel of prognostic association. The stored data labels are `prog-BL` ($c_{bm,BL}$), `prog-TV-const` ($c_{bm,TV}$), `prog-PB` ($c_{bm,PB}$), `prog-TV-grow` (b_{TV}, constant despite its label) and the block T trend pathway (b_{TV}^{tr}). The second stress test's growing natural-history pathway is the trend pathway under another name.
- **Baseline row.** An outcome row at week 0 holding the baseline level, with $D_{bc,it} = 0$ and $\eta_t = 0$. The analyses of the second stress test include it. Those of the paper 02 driver do not: their first row is the week-4 visit.
- **Feasibility bound.** The largest value of one correlation, given the others, for which the joint correlation matrix is positive definite on every path of the design.
- **Silent repair; repaired cell.** Before 2026-10-26 the drivers replaced a matrix that was not positive definite with a nearby one that was, without warning. A repaired cell is a cell simulated under such a matrix.
- **G-family.** Analyses with no expectancy terms. **G8:** $D_{bc,it}$ at the true half-life with CR2, an oracle benchmark and not a candidate analysis. **G9:** $t_{1/2}^a$ chosen by AIC on the model with the interaction, with CR2. **G10:** $t_{1/2}^a$ chosen by AIC on the model without the interaction, the interaction model refitted, with CR2. G10 is the recommended analysis of paper 02 (3). G1 and G6 are the binary-indicator analyses, with the working-covariance standard error and with CR2. Stored codes: `E2cr2`, `E7cr2`, `E7null_cr2`, `E1`, `E1cr2`.

- **F-family.** The G-family model plus η_t and $B_i^c\eta_t$, with CR2. **F1:** $t_{1/2}^a$ fixed at the half-life policy value of 0.2 week. **F1A:** chosen by AIC with the interaction in the model. **F1A0:** chosen by AIC without it. **F1A0B:** F1A0 with a single-resample parametric-bootstrap calibration that repeats the selection.
- **CR2.** The Bell-McCaffrey bias-reduced cluster-robust standard error with Satterthwaite degrees of freedom (4,5).
- **Two-MCSE band.** The interval $0.05 \pm 2\sqrt{0.05 \times 0.95/n}$ at n replicates.
- **Designs.** Hybrid: open-label titration, then blinded discontinuation, then a brief crossover. OL+BDC: open-label titration then blinded discontinuation. CO: a two-period crossover.
- **Single-resample bootstrap.** One bootstrap resample per Monte Carlo replicate, with the critical value taken from the pooled resamples of the cell.

1 Introduction

An aggregated N-of-1 trial gives each participant repeated periods on and off drug. Pooling the participants allows a test of whether a baseline biomarker modifies the drug effect. That test targets a predictive biomarker. The interaction test in a parallel-group trial already needs a large sample (6). The N-of-1 design reduces that cost by comparing each participant with themselves (7,8).

A biomarker can also be prognostic. It may track the severity at baseline, the natural course of the illness, or the response to placebo. In a design whose expectancy varies between open-label and blinded phases, the on-drug and off-drug visits differ in more than drug exposure. A prognostic association that differs between those visits can then project onto the interaction term. The test would then report moderation of the drug effect where there is none. A prognostic effect is known to bias tests of a biomarker-by-treatment interaction when the analysis model omits or misrepresents it (9,10).

The companion paper on carryover (3) studies a predictive-only biomarker. It recommends G10. That analysis uses an exposure-weighted drug indicator, chooses its half-life by AIC on the model without the interaction, and tests with CR2. This paper asks what changes when the biomarker is also prognostic. It examines five pathways. It compares the G-family with the F-family, which adds the design-expectancy terms η_t and $B_i^c\eta_t$. It also reports a finding about the simulation itself. An apparent loss of power under a strong prognostic correlation came from the silent repair of infeasible covariance matrices. It was not a property of any analysis.

Paper 01 defines the extended DGP (2). It assumes that the biomarker is not prognostic and refers that case to this paper. Two pieces of work that began there therefore appear here. The first is the joint feasibility condition. A prognostic correlation shares one feasibility condition with c_{bm}, and the extended DGP refuses a matrix that violates it (Section 3.1, Appendix B). The second is the prognostic cells of a stress test whose other cells paper 01 reports. That test fitted ten analyses, each with a baseline row, under four prognostic pathways (Section 3.6). The studies are exploratory.

Table 1: Parameter values of the published-set runs, read from the stored run metadata (`meta.rds`, field `params`). For the response components the mean column is the maximum m of the modified Gompertz curve.

Component	Maximum or mean	SD	Rate (per week)	Displacement
Natural history (TV)	6.51	10	0.35	5
Placebo response (PB)	6.51	10	0.35	5
Drug response (BR)	10.99	8	0.42	5
Biomarker	124.3	15.4	–	–
Baseline level	83.1	18.5	–	–

2 Methods

2.1 Data-generating process

All published-set cells use the extended DGP under the correlation encoding, on the Hybrid design, with $t_{1/2} = 0.5$ week unless stated. The component means and SDs are those of Hendrickson et al. Table 1 reads them from the stored run metadata. The drug-response SD is $\sigma_{BR} = 8$. The true interaction at c_{bm} is therefore $-c_{bm}\sigma_{BR}$ per SD of the biomarker: -3.60 at $c_{bm} = 0.45$ and -2.40 at $c_{bm} = 0.30$. The components are reductions in symptom severity, so an increase in any component lowers the outcome. Interaction coefficients are negative.

2.2 Prognostic pathways

A biomarker may be associated with the outcome through channels other than the drug response. Three pathways are entries in the biomarker row of the joint correlation matrix. Each sets one correlation with the others at zero. The first is a correlation with the baseline level, $c_{bm,BL}$. The second is a correlation with the natural-history component at every visit, $c_{bm,TV}$. The third is a correlation with the placebo-response component at every visit, $c_{bm,PB}$. Their strengths are bounded by feasibility (Section 3.1).

Two further pathways are mean shifts. They are not bounded by the correlation matrix. The constant shift adds $b_{TV}\,\sigma_{TV}\,z_i$ to the natural-history mean at every visit. It is stored as `prog-TV-grow` although it does not grow. It is absorbed exactly by the biomarker main effect of every analysis here (Appendix A). It serves only as a check of that absorption. The trend pathway adds $b^{\text{tr}}_{TV}\,\sigma_{TV}\,z_i\,g(w_t)$, where g is the natural-history mean course scaled to peak 1. It describes a biomarker that predicts the rate of natural-history change. A model with no biomarker-by-time term cannot absorb it.

2.3 Analyses

All analyses fit a linear mixed model with a participant random intercept and continuous-time AR(1) residual correlation (`corCAR1`). The G-family model is

$$Y_{it} = \beta_0 + \beta_B B^c_i + \beta_T w_t + \beta_D D_{bc,it} + \beta_{bm:D} B^c_i D_{bc,it} + u_i + \epsilon_{it}.$$

$D_{bc,it}$ is one on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after first exposure, and zero before first exposure. G8 sets $t^a_{1/2} = t_{1/2}$. It is an oracle benchmark and not a candidate analysis. G9 chooses $t^a_{1/2}$ by maximum-likelihood AIC over $\{0.25, 0.5, 1, 2\}$ weeks with the interaction in the model (11). G10 makes the same choice on the model without the interaction and refits the interaction model by REML. The selection of G9 depends on the coefficient under test. That is a post-selection inference risk (12,13). All three test $\beta_{bm:D}$ with CR2 and Satterthwaite degrees of freedom.

The F-family adds the design-expectancy terms:

$$Y_{it} = ... + \beta_\eta \eta_t + \beta_{bm:\eta} B^c_i \eta_t.$$

The regressors are `De` and `bmc:De`. The placebo-response amplitude scales with η_t, and η_t differs between on-drug and off-drug visits in the Hybrid design. A biomarker correlated with the placebo response therefore acquires an on/off slope difference by construction. The term $B^c_i \eta_t$ absorbs it. F1 fixes $t^a_{1/2}$ at 0.2 week. F1A selects it by AIC with the interaction. F1A0 selects it without the interaction, as G10 does. F1A0B calibrates F1A0 by a single-resample parametric bootstrap under the null that repeats the selection. Its candidate grid is $\{0, 0.2, 0.5, 1, 2\}$ weeks. η_t is constant within every path of the crossover design. The F-family is therefore not estimable under CO. F1A0 and G10 differ only in the expectancy terms. That is the property that lets this paper and paper 02 give one recommendation (Section 4.3).

2.4 Performance measures

We report the rejection rate $\hat{\pi}$ at $\alpha = 0.05$, its MCSE, the mean interaction estimate, and κ (14). Under the null $\hat{\pi}$ is the Type I error. A cell's rejection rate is compared with its two-MCSE band. That band is a check on gross departures and not a demonstration of nominal size. Within a job, cells at the same design and N share random-number streams, and every analysis is fitted to the same trials. Contrasts between analyses are paired, and their standard errors come from the per-replicate rejection indicators. Comparisons across jobs are unpaired. Results from different simulation hosts are not draw-for-draw identical.

2.5 Stored runs and their parameter sets

Table 2 lists every stored run this draft reads. The second stress test of Section 3.6 comes from another driver, the stress test of paper 01. The published-set runs are the reruns of `docs/74`. The historical runs used the simcore2026 parameter set and silent repair. They are kept only as the exhibit of Section 3.9 (`docs/74`, decision D5). No historical number enters a result about an analysis.

Table 2: Stored runs read by this draft. The parameter set is the value recorded in each run's metadata. Runs made before the parameter switch of 2026-11-03 record none and used the simcore2026 set (docs/74). The second stress test writes no metadata file; its dates span its output files, from the first run to the G10 fits. Its driver holds the published means and SDs as constants. Two analytic files of the historical folder, `pd-check.rds` and `coupling-ceilings.rds`, are read as well. The first depends on the simcore2026 SDs. The second depends on the correlation matrix only.

Run	Folder under analysis/data/	Parameter set recorded	Date	Status
Prognostic stress test	`carryover-sensitivity/published/g9-prognostic-stress`	published	2026-11-04	published set
Feasible grid and large N	`carryover-sensitivity/published/g9-gap-fill/C`	published	2026-11-05	published set
Trend pathway (block T)	`carryover-sensitivity/published/referee-variants/block-T`	published	2026-11-04	published set
Hybrid nulls (block N)	`carryover-sensitivity/published/referee-variants/block-N`	published	2026-11-04	published set
Hybrid alternatives (block A)	`carryover-sensitivity/published/referee-variants/block-A`	published	2026-11-04	published set
Original 30-cell grid	`carryover-sensitivity/published/tier1-g-vs-f-e0`	published	2026-11-04	published set
Scenario cells	`carryover-sensitivity/published/g-vs-f-comparison`	published	2026-11-04	published set
Second stress test, three runs and post hoc fits	`quick-sim/carryover-closed-form/strawman-stress`	driver constants	2026-10-07 to 2026-11-07	published set
Joint feasibility bounds	`quick-sim/carryover-closed-form/rho-stage/prognostic-joint-ceilings.csv`	correlations only	2026-10-27	analytic
Prognostic stress test	`carryover-sensitivity/g9-prognostic-stress`	not recorded	2026-10-26	historical, repaired
Random-slope variants	`carryover-sensitivity/g9-random-effects`	not recorded	2026-10-26	historical, repaired

Table 3: Largest feasible prognostic correlation given the biomarker-response correlation c_{bm}, and the reverse, under the extended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$), Hybrid design, minimum over its four paths. Correlation: the quantity whose bound is given. $c_{bm,PB}$, $c_{bm,TV}$, $c_{bm,BL}$: correlation of the biomarker with the placebo-response component, the natural-history component and the baseline level. None: no value is feasible. Computed from the correlation matrix, no simulation (`18-prognostic-joint-ceiling.R`).

Correlation	Given	$t_{1/2} = 0.2$ week	$t_{1/2} = 0.5$ week
$c_{bm,PB}$ or $c_{bm,TV}$	alone	0.513	0.513
	$c_{bm} = 0.25$	0.466	0.477
	$c_{bm} = 0.30$	0.435	0.451
	$c_{bm} = 0.45$	0.243	0.306
$c_{bm,BL}$	alone	1.000	1.000
	$c_{bm} = 0.25$	0.857	0.876
	$c_{bm} = 0.30$	0.786	0.815
	$c_{bm} = 0.45$	0.374	0.496
c_{bm}	alone	0.485	0.518
	$c_{bm,PB} = 0.3$	0.422	0.454
	$c_{bm,PB} = 0.6$	none	none

3 Results

3.1 Feasibility of a prognostic correlation

A prognostic correlation occupies the same biomarker row of the correlation matrix as c_{bm}. The two therefore share one feasibility condition. Writing the row as a sum of fixed patterns scaled by the correlations, the condition becomes a quadratic form in the correlations that must be below one. The largest value of one correlation given the others is the positive root of a quadratic (Appendix B). Table 3 gives the bounds for the Hybrid design under the extended DGP. They depend on the correlation matrix only, not on the component SDs, so they hold for both parameter sets.

At this paper's half-life of 0.5 week, a prognostic correlation of 0.3 lies inside the feasible region at every c_{bm} up to 0.45, by 0.006 at 0.45. At 0.2 week it is feasible only up to $c_{bm} =$

Table 4: Rejection rate under each prognostic pathway at strength 0.3, published parameter set. Hybrid, $t_{1/2} = 0.5$ week, 500 replicates per cell, the same trials for every analysis. Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$). G8 is the oracle benchmark.

Pathway	c_{bm}	N	G8	G9	G10	F1	F1A0	F1A0B
$c_{bm,PB}$	0.00	35	0.040	0.044	0.038	0.032	0.034	0.032
	0.00	70	0.060	0.058	0.058	0.042	0.038	0.032
	0.45	35	0.484	0.500	0.468	0.380	0.379	0.415
	0.45	70	0.786	0.792	0.780	0.658	0.678	0.646
$c_{bm,TV}$	0.00	35	0.038	0.042	0.038	0.034	0.032	0.034
	0.00	70	0.056	0.050	0.048	0.042	0.036	0.030
	0.45	35	0.418	0.431	0.402	0.371	0.377	0.405
	0.45	70	0.714	0.716	0.706	0.662	0.668	0.654
$c_{bm,BL}$	0.00	35	0.040	0.044	0.038	0.032	0.032	0.034
	0.00	70	0.056	0.054	0.048	0.042	0.038	0.032
	0.45	35	0.426	0.448	0.414	0.381	0.377	0.411
	0.45	70	0.720	0.730	0.712	0.682	0.692	0.648
b_{TV} (constant)	0.00	35	0.038	0.042	0.038	0.030	0.032	0.034
	0.00	70	0.056	0.054	0.048	0.042	0.038	0.032
	0.45	35	0.409	0.424	0.393	0.363	0.369	0.403
	0.45	70	0.720	0.724	0.710	0.694	0.698	0.680

0.42. A prognostic correlation of 0.6 with the placebo-response or natural-history component is infeasible with any c_{bm}. That is the combination that the repaired runs of Section 3.9 simulated. The correlations with the placebo-response and the natural-history components have identical bounds, because their patterns are the same constant on every visit of one component. A correlation with the baseline level has a looser bound. The baseline level is uncorrelated with the response components and enters only through its square. Outside the table, the OL+BDC bounds are within 0.034 of the Hybrid values at the same half-life. The crossover bounds are looser on c_{bm} and $c_{bm,BL}$, and slightly tighter on $c_{bm,PB}$ alone (0.487 against 0.513). Each bound was checked by the sign of the smallest eigenvalue of the full matrix: positive at 0.999 times the bound and negative at 1.001 times it. The feasible grid of Section 3.3 was chosen from these bounds.

3.2 Correlation pathways at strength 0.3

We crossed the three correlation pathways and the constant shift with $c_{bm} \in \{0, 0.45\}$ and $N \in \{35, 70\}$, at 500 replicates per cell. Table 4 gives the rejection rates. The two-MCSE band is $[0.031, 0.069]$.

The G9 Type I error was 0.042 to 0.058 across the six null cells of the three correlation pathways. G10 gave 0.038 to 0.058 and F1A0B 0.030 to 0.034. Every G9 and G10 null rate was inside the band. The G9 power at $c_{bm} = 0.45$ was 0.424 to 0.792. The no-confound reference from block A of the referee-variant runs is 0.442 at $N = 35$ and 0.728 at $N = 70$. The largest difference from it was +0.064, under the placebo-response pathway. Section 3.3 shows that this excess is bias in the estimate. That comparison is unpaired, against 1,000-replicate cells of another job. F1A0B differed from G10 in power by -0.134 to +0.010 on the same trials. The constant shift gave the same rejection at strengths 0.3 and 0.6 in every

Table 5: Block N, published parameter set: null rejection rate (MCSE) under the placebo-response pathway at $c_{bm,PB} = 0.3$, Hybrid, 5,000 replicates per cell, the same trials for every analysis, with the paired difference from G8 and its paired SE. Band $[0.044, 0.056]$.

Cell	Method	Rejection	Diff. vs G8 (SE)	κ
$t_{1/2} = 0.5$, $N = 35$	G8	0.0566 (0.0033)	–	0.95
	G9	0.0653 (0.0035)	+0.0086 (0.0017)	0.88
	G10	0.0573 (0.0033)	+0.0006 (0.0016)	0.94
	F1	0.0541 (0.0032)	-0.0028 (0.0027)	0.96
$t_{1/2} = 0.5$, $N = 70$	G8	0.0530 (0.0032)	–	1.00
	G9	0.0586 (0.0033)	+0.0056 (0.0016)	0.96
	G10	0.0538 (0.0032)	+0.0008 (0.0015)	1.00
	F1	0.0474 (0.0030)	-0.0056 (0.0025)	1.00
$t_{1/2} = 1.0$, $N = 35$	G8	0.0469 (0.0030)	–	0.97
	G9	0.0601 (0.0034)	+0.0130 (0.0021)	0.88
	G10	0.0457 (0.0030)	-0.0012 (0.0019)	0.95
	F1	0.0411 (0.0028)	-0.0060 (0.0030)	0.98
$t_{1/2} = 1.0$, $N = 70$	G8	0.0606 (0.0034)	–	0.95
	G9	0.0740 (0.0037)	+0.0134 (0.0022)	0.89
	G10	0.0622 (0.0034)	+0.0016 (0.0021)	0.93
	F1	0.0534 (0.0032)	-0.0072 (0.0033)	0.96

analysis. In the scenario job, the constant-shift null cell differs from the unshifted Hybrid null cell by at most 0.0000 in any analysis. That is the absorption of Appendix A.

A longer run at the placebo-response pathway sharpens the size comparison. Block N of the referee-variant runs fitted G8, G9, G10 and F1 to 5,000 replicates in each of 4 Hybrid null cells at $c_{bm,PB} = 0.3$ (Table 5). The band is $[0.044, 0.056]$. The paired difference from G8 was +0.0056 to +0.0134 for G9, -0.0012 to +0.0016 for G10 and -0.0072 to -0.0028 for F1. G9 was above G8 in every cell, by at least 3.5 paired standard errors. G10 was within two paired standard errors of G8 in every cell. The number of cells above the band was 2 for G8, 4 for G9, 2 for G10 and 0 for F1.

3.3 The feasible prognostic grid

A further study (driver 32, block C) ran a grid of correlations chosen inside the feasibility bounds. At $c_{bm} = 0.30$ it set $c_{bm,PB}$ and $c_{bm,TV}$ at 0.40 and 0.45, and $c_{bm,BL}$ at 0.6 and 0.8. It also ran $c_{bm,BL} = 0.45$ at $c_{bm} = 0.45$. Each cell was run with and without a true interaction, at $N = 70$ and 500 replicates. The strongest $c_{bm,PB}$ lies 0.001 below its bound at $c_{bm} = 0.30$. The strongest $c_{bm,BL}$ lies 0.015 below its bound. Table 6 gives the rejection rates and Table 7 the mean estimates.

The G9 Type I error was 0.052 to 0.066 across the 7 null cells, inside the band in every one. G10 gave 0.048 to 0.062 and F1 0.054 to 0.060. The rejection rates hide a bias that the mean estimates show. Under the placebo-response pathway with no true interaction, the mean G9 estimate was -0.54 to -0.49. G10 gave -0.52 to -0.47. F1 gave -0.09. Under the other two pathways the mean G9 estimate was -0.10 to -0.08. At $c_{bm} = 0.30$ the same bias appears as excess power. G9 reached 0.564 to 0.580 under the placebo-response pathway and 0.429 to 0.453 under the others. Its mean estimate was -2.94 to -2.87 and -2.47 to -2.45 against a true value of -2.40. F1 reached 0.380 to 0.400 under the placebo-response pathway. Its mean

Table 6: Rejection rate on the feasible prognostic grid, published parameter set. Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 500 replicates per cell. Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$). The others are power.

Cell	G8	G9	G10	F1	F1A0	F1A0B
$c_{bm,PB} = 0.40$, $c_{bm} = 0.00$	0.066	0.056	0.054	0.060	0.058	0.060
$c_{bm,PB} = 0.40$, $c_{bm} = 0.30$	0.564	0.564	0.540	0.400	0.382	0.384
$c_{bm,PB} = 0.45$, $c_{bm} = 0.00$	0.066	0.058	0.054	0.058	0.058	0.058
$c_{bm,PB} = 0.45$, $c_{bm} = 0.30$	0.582	0.580	0.566	0.380	0.390	0.370
$c_{bm,TV} = 0.40$, $c_{bm} = 0.00$	0.040	0.052	0.048	0.054	0.052	0.056
$c_{bm,TV} = 0.40$, $c_{bm} = 0.30$	0.418	0.432	0.412	0.408	0.384	0.386
$c_{bm,TV} = 0.45$, $c_{bm} = 0.00$	0.040	0.054	0.050	0.054	0.050	0.050
$c_{bm,TV} = 0.45$, $c_{bm} = 0.30$	0.439	0.438	0.410	0.406	0.388	0.392
$c_{bm,BL} = 0.45$, $c_{bm} = 0.00$	0.046	0.066	0.062	0.060	0.056	0.056
$c_{bm,BL} = 0.45$, $c_{bm} = 0.45$	0.742	0.746	0.740	0.732	0.726	0.724
$c_{bm,BL} = 0.60$, $c_{bm} = 0.00$	0.046	0.066	0.062	0.060	0.054	0.060
$c_{bm,BL} = 0.60$, $c_{bm} = 0.30$	0.432	0.453	0.427	0.399	0.383	0.391
$c_{bm,BL} = 0.80$, $c_{bm} = 0.00$	0.045	0.061	0.057	0.057	0.055	0.061
$c_{bm,BL} = 0.80$, $c_{bm} = 0.30$	0.444	0.429	0.404	0.404	0.394	0.414

Table 7: Mean interaction estimate on the feasible prognostic grid (same cells as the rejection table). The true value is $-c_{bm}\sigma_{BR}$ per SD of the biomarker.

Cell	Truth	G9	G10	F1	F1A0
$c_{bm,PB} = 0.40$, $c_{bm} = 0.00$	-0.00	-0.49	-0.47	-0.09	-0.06
$c_{bm,PB} = 0.40$, $c_{bm} = 0.30$	-2.40	-2.87	-2.78	-2.17	-2.32
$c_{bm,PB} = 0.45$, $c_{bm} = 0.00$	-0.00	-0.54	-0.52	-0.09	-0.06
$c_{bm,PB} = 0.45$, $c_{bm} = 0.30$	-2.40	-2.94	-2.84	-2.18	-2.34
$c_{bm,TV} = 0.40$, $c_{bm} = 0.00$	-0.00	-0.08	-0.09	-0.09	-0.05
$c_{bm,TV} = 0.40$, $c_{bm} = 0.30$	-2.40	-2.45	-2.39	-2.17	-2.32
$c_{bm,TV} = 0.45$, $c_{bm} = 0.00$	-0.00	-0.08	-0.08	-0.08	-0.04
$c_{bm,TV} = 0.45$, $c_{bm} = 0.30$	-2.40	-2.47	-2.41	-2.19	-2.35
$c_{bm,BL} = 0.45$, $c_{bm} = 0.00$	-0.00	-0.10	-0.09	-0.09	-0.06
$c_{bm,BL} = 0.45$, $c_{bm} = 0.45$	-3.60	-3.64	-3.56	-3.22	-3.47
$c_{bm,BL} = 0.60$, $c_{bm} = 0.00$	-0.00	-0.09	-0.09	-0.09	-0.07
$c_{bm,BL} = 0.60$, $c_{bm} = 0.30$	-2.40	-2.46	-2.39	-2.17	-2.31
$c_{bm,BL} = 0.80$, $c_{bm} = 0.00$	-0.00	-0.09	-0.09	-0.11	-0.08
$c_{bm,BL} = 0.80$, $c_{bm} = 0.30$	-2.40	-2.47	-2.41	-2.18	-2.34

Table 8: Type I error under the placebo-response pathway at larger samples, published parameter set. Hybrid, $t_{1/2} = 0.5$ week, $c_{bm} = 0$, 1,000 replicates at $N = 140$ and 280 (band $[0.036, 0.064]$). The $N = 70$ row is from the feasible grid at 500 replicates.

Cell	G1	G8	G9	G10	F1	F1A0	F1A0B
$c_{bm,PB} = 0.30$, $N = 140$	0.048	0.060	0.062	0.058	0.052	0.050	0.041
$c_{bm,PB} = 0.30$, $N = 280$	0.055	0.065	0.067	0.066	0.056	0.053	0.050
$c_{bm,PB} = 0.45$, $N = 70$	0.046	0.066	0.058	0.054	0.058	0.058	0.058
$c_{bm,PB} = 0.45$, $N = 140$	0.057	0.076	0.077	0.073	0.051	0.049	0.042
$c_{bm,PB} = 0.45$, $N = 280$	0.069	0.094	0.092	0.089	0.060	0.058	0.053

estimate was -2.19 to -2.17 across the grid. The F1 estimate is attenuated because its fixed half-life of 0.2 week is below the true 0.5 week (Section 3.8). The excess power of the G-family under the placebo-response pathway is therefore not evidence of better power. It is the same bias as under the null.

3.4 Larger samples under the placebo-response pathway

A bias of a fixed size becomes Type I error inflation as N grows, because the standard error shrinks. The same study ran the placebo-response pathway with no true interaction at $N \in \{140, 280\}$ and $c_{bm,PB} \in \{0.30, 0.45\}$, at 1,000 replicates. The band is $[0.036, 0.064]$. Table 8 gives the result.

At $N = 280$ and $c_{bm,PB} = 0.45$, G9 rejected at 0.092, G10 at 0.089 and G8 at 0.094. The binary-indicator analysis with the working-covariance standard error, G1, rejected at 0.069. The F-family stayed inside the band: F1 at 0.060, F1A0 at 0.058 and F1A0B at 0.053. On the same trials, the paired difference in rejection of F1 minus G9 was -0.032 (paired SE 0.009). That of F1A0 minus G10 was -0.031 (0.009). The G9 rate rose with N: 0.058 at 70, 0.077 at 140 and 0.092 at 280. Its mean estimate at $N = 280$ was -0.36 against +0.05 for F1. The standard errors were calibrated for both, with κ of 0.96 and 0.97. The excess is therefore bias, not an underestimated standard error. At $c_{bm,PB} = 0.30$ the rates were 0.067 for G9 and 0.056 for F1.

3.5 The trend pathway

Block T of the referee-variant runs simulated a natural-history shift that grows over the trial, with $b_{TV}^{\mathrm{tr}} = 0.3$. It is a mean shift, so the feasibility bound does not limit it. G8, G9 and F1 were fitted at $N = 70$ and 1,000 replicates (Table 9). The band is $[0.036, 0.064]$.

The trend pathway biased the G-family. Under the null G8 rejected at 0.063 and G9 at 0.073. Their mean estimates were +0.43 and +0.42 where the truth is zero. The G8 empirical SE was 1.45. F1 rejected at 0.052, with a mean estimate of +0.24. Paired against G8, F1 differed by -0.011 (SE 0.006) and G9 by +0.010 (0.005). The expectancy terms therefore absorb part of the shift but not all of it. At $c_{bm} = 0.45$ the bias is toward zero. G9 power was 0.631 with a mean estimate of -3.12. In the no-trend cell of block A it was 0.728 with -3.47. That

Table 9: Block T, published parameter set: rejection rate (MCSE), mean interaction estimate, empirical SE and calibration ratio under the trend pathway ($b_{TV}^{\text{tr}} = 0.3$). Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 1,000 replicates. The true value is 0 at the null and -3.60 at $c_{bm} = 0.45$. Band at the null $[0.036, 0.064]$.

Cell	Method	Rejection	Mean estimate	Empirical SE	κ
null ($c_{bm} = 0$)	G8	0.063 (0.008)	+0.43	1.45	0.97
	G9	0.073 (0.008)	+0.42	1.46	0.93
	F1	0.052 (0.007)	+0.24	1.27	0.98
$c_{bm} = 0.45$	G8	0.629 (0.015)	-3.20	1.37	0.97
	G9	0.631 (0.015)	-3.12	1.41	0.90
	F1	0.632 (0.015)	-2.88	1.23	0.97

comparison is unpaired. The mechanism is the design. In the Hybrid design $D_{bc,it}$ falls with calendar time, so a biomarker that predicts the trend projects onto the interaction term of a model with no biomarker-by-time term. The paper 02 driver has no baseline row. The second stress test applied the same shift to analyses with a baseline row and found a bias of the opposite sign (Section 3.6.1).

3.6 A second stress test: ten analyses under the extended DGP

This section reports the prognostic cells of a second stress test, run with the driver of paper 01 (`analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R`). Paper 01 reports the cells of the same study, its study 2, that have no prognostic pathway (2). The trials come from the extended DGP with the published parameter set, in the Hybrid design: $N = 70$ with fixed allocation (18, 18, 17, 17), $t_{1/2} = 0.5$ week, $\rho = 0.7$ and $c_1 = 0.2$. Four null cells ($c_{bm} = 0$) each add one prognostic pathway of strength 0.3: $c_{bm,TV} = 0.3$, $c_{bm,PB} = 0.3$, $c_{bm,BL} = 0.3$, and a mean shift of the natural-history component of 0.3 of its SD per SD of the biomarker at the end of its curve, growing along the curve. The last is labeled the growing natural-history pathway here. It is not the stored `prog-TV-grow` cell of the paper 02 driver, which is constant (Section 2.2). It is the trend pathway of Section 3.5, with the same coefficient (Section 3.6.1). Each cell has 1,000 replicates, so the single-cell two-MCSE band is 0.036 to 0.064.

Fourteen analyses were fitted to the same trials, and ten are reported. They are named by their model in Table 10, not by the G and F codes of this paper. Every one of them keeps the baseline visit as an outcome row. The expectancy-term model adds η_t and $B_i^c \eta_t$, as the F-family does. The free-baseline model adds `bmc:bl0`, a separate biomarker slope for the baseline row. Analyses with a fixed analysis half-life use 0.5 week, the true value.

The study was run three times with the same seed rule, $20261007 + 1000 \times \text{cell} + \text{chunk}$. The first run used the compounding carryover rule on the mean and fixed allocation. The second used carryover decaying from the last on-drug mean and randomized allocation. The third, reported here, is the extended DGP as defined: decaying carryover and fixed allocation. It differs from the first only in the carryover rule and from the second only in the allocation. Failed fits were excluded from the denominator. They were at most 0.6% in any cell. The

Table 10: Second stress test: null rejection rates under a prognostic biomarker ($c_{bm} = 0$). Each pathway has strength 0.3: a correlation of 0.3 with the natural-history component (constant), the placebo-response component or baseline severity, and for the growing natural-history pathway a mean shift of the natural-history component of 0.3 of its SD (3 points) per SD of biomarker at the end of its curve, growing along the curve. 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Third run: the extended DGP, with carryover decaying from the last on-drug mean and fixed path allocation. Every analysis has a baseline row. Rates above 0.075 in bold. The first run is in Appendix D.

Analysis	Natural history, constant	Natural history, growing	Placebo response	Baseline severity
Binary-indicator analysis (random intercept, model-based)	**0.108**	0.056	**0.157**	0.064
Random intercept + CAR(1), CR2	**0.103**	0.049	**0.105**	0.052
The same, half-life chosen by AIC	**0.130**	0.059	**0.125**	0.057
Unstructured covariance, Kenward-Roger	**0.413**	**0.200**	**0.224**	0.048
The same, half-life chosen by AIC	**0.506**	**0.258**	**0.270**	0.054
Random intercept + CAR(1) with expectancy terms, CR2	0.045	0.040	0.052	0.055
The same, with phase-specific residual variances, CR2	0.056	0.037	0.051	0.059
The same, with phase-specific residual variances, model-based	0.054	0.042	0.049	0.059
Expectancy terms, half-life chosen by AIC, CR2	0.058	0.050	0.060	0.060
Random intercept + CAR(1) with free baseline slope, CR2	0.051	0.066	0.060	0.054

study is exploratory. Its cells and analyses followed a working protocol that was revised as results came in. Two cells with a prognostic association with baseline severity, and three analyses, were added after the first results.

Analyses that keep baseline as an outcome row use one biomarker main effect for baseline and for the off-drug visits. When the biomarker is prognostic through a pathway that is absent at baseline, those two associations differ, and part of the difference is attributed to the interaction (Appendix C). Every analysis that shares the biomarker effect with baseline exceeded the nominal level in at least one such cell (Table 10).

The expectancy-term family stayed within the single-cell two-MCSE band in every prognostic cell. Its four analyses rejected 0.037 to 0.060. This family adds the expectancy and a biomarker-by-expectancy term to the mean model, with or without residual variances that differ by trial phase. The free-baseline model rejected 0.066 under the growing natural-history pathway, above the single-cell band, and 0.060 under the placebo pathway. Appendix C explains why freeing the baseline slope alone is not sufficient when the prognostic slope grows over time or scales with the expectancy. A prognostic association with baseline severity alone did less harm, because baseline severity enters every visit equally. The binary-indicator analysis rejected 0.064 there. The scenarios in which the pathway is absent at baseline are the ones that show the inflation. Whether such a pathway, at a magnitude of 0.3, is realistic for

blood pressure in PTSD is not known.

Changes between runs. Treating two runs as independent, the difference between them in one cell has a Monte Carlo SE of about 0.010 at a rate of 0.05. The first and third runs use the same seeds and allocation, so their differences are paired and that SE overstates their noise. The carryover rule changed the table by at most 0.017, for the expectancy-term model with phase-specific variances (CR2) under the baseline-severity pathway (Appendix D). Cells that crossed the bold threshold of 0.075: none. Cells that crossed the upper edge of the single-cell band: the free-baseline model under the growing natural-history pathway (first run 0.063, third run 0.066); the unstructured model with AIC-chosen half-life under the baseline-severity pathway (first run 0.064, third run 0.054). Randomized allocation changed it by at most 0.041, for the unstructured model with AIC-chosen half-life under the placebo pathway. Cells that crossed the bold threshold: the binary-indicator analysis under the growing natural-history pathway (second run 0.077, third run 0.056). Cells that crossed the upper edge of the single-cell band: the binary-indicator analysis under the growing natural-history pathway (second run 0.077, third run 0.056); the CAR(1) model with AIC-chosen half-life under the growing natural-history pathway (second run 0.068, third run 0.059); the unstructured model with AIC-chosen half-life under the baseline-severity pathway (second run 0.065, third run 0.054). In all three runs the expectancy-term family stayed within the single-cell band in every prognostic cell. The binary-indicator, CAR(1) and unstructured analyses each exceeded it in at least one cell, with or without an AIC-chosen half-life. Taken analysis by analysis, only the free-baseline model changed its classification. It stayed within the band in the first run and exceeded it under the growing pathway in the other two, by at most 0.009.

G10 under these pathways. The planning sample sizes for G10 (paper 02) assume that the biomarker is not prognostic. We fitted G10 to the trials of this stress test, with the baseline row. It rejected the null at 0.052 in the base cell, and at 0.108, 0.051, 0.110 and 0.050 under the constant natural-history, growing natural-history, placebo and baseline-severity pathways (single-cell band 0.036 to 0.064). The expectancy-term model rejected 0.040 to 0.055 in the same cells. In this test G10 held its level when the biomarker acted through baseline severity or the growing natural history, but not when it acted through a constant natural history or the placebo response. Section 3.6.1 shows why the growing case may not carry over to an analysis without a baseline row. A trial in which a constant natural-history or placebo pathway is plausible should be planned for the expectancy-term analysis. On the power-curve trials of the stress test, matching G10's power with that analysis needs about 1.13 to 1.27 times the G10 sample size, by the normal approximation to power, at c_{bm} from 0.225 to 0.45. At $c_{bm} = 0.15$ both analyses have power near 0.15 and the factor, 0.96, is within Monte Carlo error of 1. Paper 02's G10 sample sizes can be scaled by that factor. The factor comes from one design, half-life and ρ, the stress test's base cell.

Power and robustness. The expectancy-term model keeps its size under every prognostic pathway and has lower power. In the base cell at $c_{bm} = 0.30$ its power was 0.404 against 0.486 for the CAR(1) model. A further 1,500 trials per cell under the second run's DGP (randomized allocation) were pooled with that run (2,000 per cell). There, size-adjusted power was 0.431 against 0.496, about 13% less. Under compound symmetry at $c_{bm} = 0.25$ the gap closes.

Table 11: The growing natural-history shift ($0.3\,\sigma_{TV}\,z_i\,g(w_t)$) with no true interaction, in the two drivers. Each entry: rejection rate (mean interaction estimate per SD of the biomarker; empirical SE). Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, fixed allocation, 1,000 replicates. Band $[0.036, 0.064]$. The estimates of the second test are converted from raw biomarker units by its SD.

Analysis	Block T, no baseline row	Second stress test, baseline row
Fixed half-life 0.5 week, CR2	G8: 0.063 (+0.43; 1.45)	CAR(1): 0.049 (-0.14; 1.22)
Half-life by AIC, interaction in the model	G9: 0.073 (+0.42; 1.46)	AIC variant: 0.059 (-0.19; 1.28)
Half-life by AIC, interaction not in the model	not fitted	G10: 0.051 (-0.18; 1.24)
Expectancy terms, fixed half-life	F1 (0.2 week): 0.052 (+0.24; 1.27)	Expectancy-term (0.5 week): 0.040 (+0.03; 1.37)

Power was 0.564 against 0.552 in the reported run, and size-adjusted power in the pooled trials was 0.542 against 0.539 (Monte Carlo SE about 0.011). The choice between power and robustness is the trialist's. It depends on whether a prognostic pathway absent at baseline is plausible for the biomarker in hand.

3.6.1 The growing pathway and the trend pathway

The growing natural-history pathway of this test and the trend pathway of Section 3.5 are the same mean shift. Both add $0.3\,\sigma_{TV}\,z_i\,g(w_t)$ to the natural-history component, with g the natural-history mean course scaled to peak 1 over the visits. Driver 11 applies it in `draw()`. The paper 02 driver applies it in `implementations/tidyverse/R/functions.R`. Both use the published natural-history curve, $\sigma_{TV} = 10$, the Hybrid design with fixed allocation, $N = 70$, $t_{1/2} = 0.5$ week and 1,000 null replicates. The two results appear to disagree. Under the trend pathway G9 exceeded the band. Under the growing pathway G10 stayed inside it.

The comparison is less direct than that. The two tests did not fit the same analyses. Block T fitted G8, G9 and F1, not G10. The second test fitted G10 but not G8 or G9. Its CAR(1) model with CR2 at the fixed half-life of 0.5 week has the formula, covariance and test of G8. Its AIC variant resembles G9, with a grid that adds 0. The two tests also differ in their analysis data. Every analysis of the second test has a baseline row. The paper 02 driver has none. Table 11 sets the comparable analyses side by side. The comparison is across jobs and unpaired.

At the fixed half-life the two drivers agree on size. G8 rejected at 0.063 and the CAR(1) model at 0.049, both inside the band. G9 rejected at 0.073, above it, and the AIC variant of the second test at 0.059. The mean estimates differ in sign. Without a baseline row G8 gave +0.43 per SD of the biomarker. With one, the CAR(1) model gave -0.14 and G10 -0.18. Every one of these is within a third of its empirical SE of zero.

The baseline row accounts for the change of sign. The shift is zero at week 0 and grows along the natural-history curve. In the Hybrid design the off-drug visits come late, where the shift is large. Without a baseline row the off-drug visits carry more of the shift than the on-drug visits. The shift lowers the outcome, so the difference projects onto $B_i^c D_{bc,it}$ with a

positive sign. A baseline row adds an off-drug row with no shift. That reverses the sign. An unweighted least-squares projection of the shift onto the biomarker terms of the G8 model gives +0.69 per SD of the biomarker without the baseline row and -0.08 with it. It ignores the random intercept and the CAR(1) weighting, so it checks only the signs, which agree with the fitted estimates.

The two results therefore do not conflict. The DGP settings match. The difference lies in the analysis data and in which analyses were fitted. Two points remain open. First, G10 has not been fitted under the trend pathway without a baseline row. That it held its level under the growing pathway rests on analyses with a baseline row only. Second, the bias is real in both tests. Every G-type mean estimate in Table 11 is at least 3.6 Monte Carlo SEs from zero. It is within a third of the empirical SE at $N = 70$, so it moves the rejection rate little there. As under the placebo-response pathway (Section 3.4), a fixed bias of either sign inflates the size of the two-sided test as N grows. The sign here depends on the visit structure of the analysis. A larger sample under the trend pathway, with the same analysis fitted with and without the baseline row, would settle both points. The biomarker-by-time block of Section 3.7 should include that contrast.

3.7 The biomarker-by-time remedy: RESULTS PENDING

RESULTS PENDING. The simulation for this section has not been run. This section contains no results. It will be written from the block described in the split plan, Section 3.1, once that block is stored.

The remedy adds a biomarker-by-time term $B_i^c w_t$ to the analysis model, as `bmc:t`. Its coefficient would be written $\beta_{bm:T}$ (*proposed*). A quadratic variant adds $B_i^c w_t^2$. The planned block crosses the trend pathway, present or absent, with a null and an alternative interaction. It fits G8, G9, G10 and F1A0 with and without the term to the same trials. It answers three questions. Does the term remove the trend bias in size and in the mean estimate? What does it cost in power when there is no trend? Does a linear term suffice when the natural-history course is not a straight line? Section 3.6.1 adds a fourth: does the answer depend on whether the analysis has a baseline row? The block should fit each analysis both ways. The driver changes it needs are listed in the Reproducibility section.

3.8 Expectancy terms without a prognostic effect

The F-family's cost when the biomarker is predictive only was measured on the original 30-cell grid of paper 02 under the published parameter set (500 replicates per cell). The F-family is not estimable under CO, so only the Hybrid and OL+BDC cells enter. Table 12 gives the rejection rates.

The null rates were 0.044 to 0.062 for G10, 0.054 to 0.056 for F1A0, 0.040 to 0.062 for F1A0B and 0.052 to 0.064 for F1. Over the alternative cells of the Hybrid design, F1A0 reached 98.2 percent of the mean power of G10, F1A0B 97.4 percent and F1 95.5 percent. Under OL+BDC the figures were 93.3, 86.3 and 88.0 percent. The mean paired difference from G10 over the

Table 12: Original 30-cell grid of paper 02, published parameter set, Hybrid and OL+BDC cells: rejection rate with no prognostic effect. 500 replicates per cell, the same trials for every analysis. Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$).

Design	c_{bm}	$t_{1/2}$	N	G9	G10	F1	F1A0	F1A0B
Hybrid	0.00	1.0	35	0.078	0.060	0.054	0.056	0.062
	0.00	1.0	70	0.066	0.062	0.052	0.054	0.054
	0.30	1.0	35	0.179	0.155	0.136	0.140	0.154
	0.30	1.0	70	0.327	0.299	0.280	0.280	0.272
	0.45	0.0	35	0.512	0.508	0.486	0.486	0.486
	0.45	0.0	70	0.868	0.868	0.872	0.874	0.874
	0.45	0.5	35	0.394	0.376	0.362	0.374	0.360
	0.45	0.5	70	0.749	0.737	0.718	0.711	0.699
	0.45	1.0	35	0.317	0.277	0.248	0.279	0.287
	0.45	1.0	70	0.597	0.553	0.501	0.561	0.543
OL+BDC	0.00	1.0	35	0.052	0.044	0.064	0.056	0.048
	0.00	1.0	70	0.056	0.050	0.060	0.056	0.040
	0.30	1.0	35	0.114	0.108	0.086	0.106	0.094
	0.30	1.0	70	0.204	0.198	0.150	0.192	0.140
	0.45	0.0	35	0.401	0.390	0.382	0.372	0.384
	0.45	0.0	70	0.721	0.715	0.695	0.682	0.644
	0.45	0.5	35	0.305	0.289	0.220	0.258	0.270
	0.45	0.5	70	0.567	0.563	0.510	0.508	0.454
	0.45	1.0	35	0.202	0.198	0.142	0.178	0.158
	0.45	1.0	70	0.385	0.365	0.300	0.340	0.296

8 Hybrid alternative cells was -0.009 for F1A0 (SE 0.004) and -0.022 for F1 (0.005). Under OL+BDC it was -0.024 (0.005) and -0.043 (0.005). The pooled standard errors treat the cells as independent. Cells at the same N share random-number streams, so they are approximate.

F1 also loses power where its fixed half-life is wrong. In block A at a true half-life of 2 weeks and $c_{bm} = 0.45$, F1 reached 0.163 to 0.279 against 0.192 to 0.359 for G10. Its mean estimate was -1.85 to -1.72 against -3.09 to -2.93 for G10 and a true value of -3.60. The fixed half-life is the cost, not the expectancy terms. F1A0, which selects the half-life, is the fair comparator for the terms themselves.

Table 13: Scenario cells of paper 02, published parameter set, no prognostic effect: rejection rate. Hybrid unless stated, $N = 70$, $t_{1/2} = 0.5$ week unless stated, 500 replicates. In the $t_{1/2} = 2$ week rows the fixed half-life of F1 is a tenth of the true value.

Cell	G9	G10	F1	F1A0	F1A0B
Hybrid, null	0.060	0.052	0.068	0.058	0.056
OL+BDC, null	0.060	0.058	0.048	0.048	0.046
$t_{1/2} = 2$ wk, null	0.078	0.066	0.068	0.058	0.052
Weibull $k = 2$ data, null	0.062	0.058	0.068	0.064	0.054
Hybrid, $c_{bm} = 0.45$	0.754	0.736	0.728	0.722	0.710
Hybrid, $c_{bm} = 0.25$	0.296	0.288	0.290	0.282	0.280
OL+BDC, $c_{bm} = 0.45$	0.576	0.566	0.524	0.532	0.480
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.162	0.138	0.104	0.142	0.132
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.378	0.374	0.362	0.362	0.332

Table 14: Historical exhibit (simcore2026 parameter set, silent repair): power at $c_{bm} = 0.45$ at prognostic strengths 0.3 and 0.6, Hybrid, $t_{1/2} = 0.5$ week, 500 replicates. Every strength-0.6 cell here, including $c_{bm,BL} = 0.6$, was simulated under a repaired matrix.

		G9		F1		F1A0B	
Pathway	N	0.3	0.6	0.3	0.6	0.3	0.6
$c_{bm,PB}$	35	0.798	0.597	0.728	0.524	0.741	0.562
	70	0.968	0.853	0.946	0.798	0.940	0.794
$c_{bm,TV}$	35	0.782	0.502	0.720	0.454	0.730	0.482
	70	0.968	0.772	0.952	0.718	0.950	0.702
$c_{bm,BL}$	35	0.782	0.738	0.713	0.676	0.758	0.702
	70	0.976	0.950	0.958	0.929	0.960	0.920

3.9 The repair artifact: a historical exhibit

Historical exhibit. Every number in this section comes from runs under the simcore2026 parameter set with silent repair (Table 2). They are evidence about a repaired DGP. They are not evidence about any analysis under a prognostic biomarker.

The first prognostic stress test ran at strengths 0.3 and 0.6 before the drivers refused infeasible matrices. At strength 0.6 on the placebo-response and natural-history pathways, G9 lost 12 to 26 points of power at $c_{bm} = 0.45$ against the no-confound reference of 0.759 ($N = 35$) and 0.969 ($N = 70$). At strength 0.3 the largest loss was 0.039. Every one of the 14 analyses fitted to the same trials lost power at strength 0.6 (Table 14). That uniformity was the first sign that the loss arose before the analysis.

A direct check of the covariance matrices explains it (Table 15). Each strength-0.6 cell on those two pathways, and $c_{bm,BL} = 0.6$ at $c_{bm} = 0.45$, has a joint matrix that is not positive definite on all 4 Hybrid paths. The repair replaced it with a nearby matrix before any data were drawn. The repair shrank the population on-drug minus off-drug slope of the outcome on the biomarker. That is the quantity the interaction term estimates. With no prognostic correlation it is -3.26. After repair it was -2.38 under the placebo-response pathway and -1.81 under the natural-history pathway. The power loss is the repair removing part of the interaction from the data. No analysis could recover it. At strength 0.3 the matrix is positive definite, with a smallest eigenvalue of 0.010 under the placebo-response pathway.

Before the artifact was found, the loss was read as patient-level variance that a random intercept could not absorb. G9 was refitted with a random slope on time, on expectancy, and on both, in 12 cells at 1,000 replicates. 8 of those cells are repaired strength-0.6 cells. The three variants matched plain G9 to within 0.010 in rejection rate in every cell (Table 16). The largest paired per-replicate difference was 0.008. Non-convergence reached 2.5 percent for the variant with both slopes. The DGP has no patient-specific slope, so there was no variance for a random slope to absorb. The check shows only that the slopes cost nothing there.

The published-set rerun applied the current default, which refuses an infeasible matrix. Of its 12 strength-0.6 cells on the three correlation pathways, 10 were refused and 2 ran. The refused cells are exactly the configurations that the historical check found infeasible. The cells that ran set $c_{bm,BL} = 0.6$ with no true interaction, which is feasible. Under the current code

Table 15: Historical exhibit: positive definiteness of the raw joint covariance matrix on the Hybrid paths, and the population on-drug minus off-drug slope of the outcome on the biomarker before and after the silent repair. Computed from the matrix under the simcore2026 SDs, no simulation (`30-check-prognostic-confound-pd.R`).

Pathway	Strength	c_{bm}	Paths repaired	Min. eigenvalue	Contrast, raw	Contrast, repaired
none	0.0	0.45	0	0.180	-3.26	-3.26
$c_{bm,BL}$	0.3	0.45	0	0.145	-3.26	-3.26
$c_{bm,BL}$	0.6	0.00	0	0.184	+0.00	+0.00
$c_{bm,BL}$	0.6	0.45	4	-0.110	-3.26	-3.00
$c_{bm,PB}$	0.3	0.45	0	0.010	-3.40	-3.40
$c_{bm,PB}$	0.6	0.00	4	-0.240	-0.27	-0.23
$c_{bm,PB}$	0.6	0.45	4	-0.642	-3.53	-2.38
$c_{bm,TV}$	0.3	0.45	0	0.011	-3.26	-3.26
$c_{bm,TV}$	0.6	0.00	4	-0.354	+0.00	+0.00
$c_{bm,TV}$	0.6	0.45	4	-0.853	-3.26	-1.81

Table 16: Historical exhibit: rejection rate of G9 and its three random-slope variants (simcore2026 parameter set, Hybrid, $t_{1/2} = 0.5$ week, 1,000 replicates). The strength-0.6 rows were simulated under repaired matrices.

Cell	G9	+ slope on w_t	+ slope on η_t	+ both
none, $c_{bm} = 0.00$, $N = 35$	0.050	0.050	0.050	0.046
none, $c_{bm} = 0.00$, $N = 70$	0.043	0.044	0.043	0.041
none, $c_{bm} = 0.45$, $N = 35$	0.759	0.762	0.766	0.769
none, $c_{bm} = 0.45$, $N = 70$	0.969	0.970	0.969	0.970
$c_{bm,PB}$ 0.6, $c_{bm} = 0.00$, $N = 35$	0.060	0.060	0.059	0.060
$c_{bm,PB}$ 0.6, $c_{bm} = 0.00$, $N = 70$	0.047	0.046	0.047	0.047
$c_{bm,PB}$ 0.6, $c_{bm} = 0.45$, $N = 35$	0.594	0.593	0.593	0.596
$c_{bm,PB}$ 0.6, $c_{bm} = 0.45$, $N = 70$	0.860	0.859	0.858	0.858
$c_{bm,TV}$ 0.6, $c_{bm} = 0.00$, $N = 35$	0.050	0.053	0.051	0.054
$c_{bm,TV}$ 0.6, $c_{bm} = 0.00$, $N = 70$	0.044	0.044	0.043	0.040
$c_{bm,TV}$ 0.6, $c_{bm} = 0.45$, $N = 35$	0.489	0.488	0.489	0.485
$c_{bm,TV}$ 0.6, $c_{bm} = 0.45$, $N = 70$	0.772	0.772	0.773	0.768

a repaired cell cannot be simulated without an explicit option.

4 Discussion

4.1 What the expectancy terms are for

The placebo-response amplitude scales with η_t. In the Hybrid design η_t differs between on-drug and off-drug visits. A biomarker correlated with the placebo response therefore acquires an on/off slope difference with no true interaction. Under the simcore2026 SDs the population contrast was -0.13 at $c_{bm,PB} = 0.3$ and -0.27 at 0.6, against -3.26 for the interaction at $c_{bm} = 0.45$ (historical, Table 15). The placebo-response SD is 2 in the simcore2026 set and 10 in the published set. The contrast scales with that SD. It has not been recomputed under the published set.

The published-set runs show the consequence. Under the placebo-response pathway the G-family estimate moved away from zero with no interaction present (Section 3.3). The bias was small against the standard error at $N = 70$ and produced no excess rejection there. At $N = 280$ it did (Section 3.4). The F-family, whose $B_i^c \eta_t$ term absorbs the contrast, stayed inside the band at every N examined. The earlier conclusion that the expectancy terms add nothing came from the simcore2026 runs. It does not hold under the published parameter values.

4.2 Whether the analysis has a baseline row

The results come from two kinds of analysis data. The paper 02 driver starts at the first post-baseline visit. The second stress test adds baseline as an off-drug row. The choice changes how a prognostic pathway reaches the interaction term. Without a baseline row, the correlation pathways at strength 0.3 left G9 and G10 inside the band at $N \leq 70$ (Section 3.2), and the placebo-response bias became visible only at $N = 280$ (Section 3.4). With a baseline row, a pathway absent at baseline also enters through the single biomarker main effect shared by baseline and the off-drug visits (Appendix C). G10 then exceeded the band at $N = 70$ under the natural-history and placebo-response correlations (Section 3.6). Under the trend pathway the baseline row reversed the sign of the bias (Section 3.6.1). The expectancy-term analyses kept their size in both kinds of data. Which convention a trial's analysis follows is therefore part of its specification. A study that plans G10 for a biomarker that may be prognostic should state it.

4.3 The joint recommendation

This paper and paper 02 give one recommendation between them.

- For a biomarker believed to be predictive only, use G10: the exposure-weighted indicator with the half-life chosen by AIC on the model without the interaction, tested with CR2

(3).

- Where the biomarker may also be prognostic, add the design-expectancy terms η_t and $B_i^c \eta_t$. This is F1A0, or its bootstrap-calibrated form F1A0B. F1A0 and G10 differ only in those two terms.
- Where the biomarker may predict the rate of natural-history change, also add a biomarker-by-time term. This step is provisional until the block of Section 3.7 is run.

F1, which fixes the half-life at 0.2 week, is not recommended. It loses power wherever that value is wrong (Section 3.8). Two questions remain for F1A0B. Its bootstrap calibration may be unnecessary once the selection excludes the interaction. Its cost against G10 without a prognostic effect is measured here on one grid only.

4.4 Limitations

The published-set results come from one parameter set and one correlation structure ($\rho = 0.7$, $c_1 = 0.2$). Most cells are on the Hybrid design at $t_{1/2} = 0.5$ week. The prognostic correlations are correlations of a single latent biomarker with the components. The feasibility bound is a property of that parameterization. The large-sample cells have no confound-free counterpart at the same N. The trend pathway was simulated at one coefficient, one sample size and one half-life. The sign of its bias depends on whether the analysis has a baseline row, and no single job has fitted the same analysis both ways. G10 has not been fitted under it without a baseline row. Its remedy has not been simulated. The second stress test ran 1,000 null replicates per cell at one sample size, and its cells and analyses were revised as results came in. The placebo-response contrast has not been recomputed under the published SDs. The random-slope variants were fitted only to historical, partly repaired cells under a DGP without patient-specific slopes. Comparisons with the block A reference are across jobs and unpaired. Cells within a job share random-number streams, so counts of cells inside a band are not independent confirmations.

5 Conclusions

1. A constant prognostic shift is absorbed exactly by the biomarker main effect. It cannot move the interaction estimate.
2. Inside the feasibility bounds, and without a baseline row, the correlation pathways left the G9 and G10 Type I error inside the band at $N \leq 70$.
3. A biomarker correlated with the placebo response biases the G-family estimate. Under the published parameter values the bias inflated the size of G9 and G10 at $N = 280$. The expectancy terms removed it.
4. A biomarker that predicts the natural-history trend biased every analysis fitted by the paper 02 driver, which has no baseline row. The expectancy terms absorbed part of the bias. With a baseline row the bias changed sign. At $N = 70$ the mean estimate was within a third of its empirical SE of zero in both. The biomarker-by-time remedy is pending.

5. In the second stress test, whose analyses have a baseline row, the expectancy-term analyses kept their size under all four pathways. G10 exceeded the band under the constant natural-history and placebo-response pathways.
6. The power loss once attributed to a prognostic correlation of 0.6 was produced by the silent repair of infeasible matrices. Under the current default those cells are refused.
7. For a biomarker that may be prognostic, add η_t and $B_i^c \eta_t$ to G10. The result is F1A0.

Declarations

Funding, competing interests, author contributions. To be completed.

Data and code availability. Every number in this draft is read from the stored files of Table 2 by inline code.

6 Reproducibility

The drivers live in `analysis/scripts/carryover-sensitivity/`. The published-set runs were submitted on 2026-11-03 under `docs/74`, wave 1. The prognostic stress test is driver 28. The feasible grid and the large-sample cells are driver 32, block C. The trend, null and alternative blocks are driver 36, blocks T, N and A. The original grid is driver 27 and the scenario cells driver 06. Driver 29, the random-slope variants, was not rerun. Every confounded cell in it is a repaired strength-0.6 cell (`docs/74`, decision D5).

The second stress test is the stress test of paper 01, which reports its cells without a prognostic pathway. Its driver is `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R`, and its outputs are under `analysis/data/quick-sim/carryover-closed-form/strawman-stress/`. The driver holds the published means and SDs as constants and stops on a matrix that is not positive definite. The reported (third) run is reproduced with `--carryover anchored --allocation fixed --analysis-half-life 0.5` (files tagged `-anchored-fixed-ha0.5`). The second run used `--carryover anchored --allocation random --analysis-half-life 0.5` (tagged `-anchored`). The first used `--carryover recursive --allocation fixed --analysis-half-life 0.5` (untagged). Each run was followed by the same command with `--extra` and with `--extra2` added, on the same trials. G10 was fitted to the trials of the third run by the same command with `--extra5`. The pooled size-adjusted comparison adds 1,500 fresh trials per cell from `--extra4` under the second run's DGP (tagged `-extra4-anchored`); `15-extra4-pooled-inputs.R` pools them with the second run and writes `strawman-stress/extra4-decision/pooled/`. The script's default analysis half-life is now 0.2 week, so the `--analysis-half-life 0.5` flag is needed. The joint feasibility bounds of Table 3 come from `analysis/scripts/quick-sim/carryover-closed-form/18-prognostic-joi` which writes `analysis/data/quick-sim/carryover-closed-form/rho-stage/prognostic-joint-ceilings.` It computes the bounds from the correlation matrix, with no simulation.

Changes needed before the next draft, not made here:

- Add the biomarker-by-time specifications (`E2cr2_bmt`, `E7cr2_bmt`, `E7null_cr2_bmt`, `F1A0_bmt`, optionally quadratic) to `simulation-core.R`. Add the corresponding block to `36-run-referee-variants.R` (split plan, Section 3.1, step 6).
- Give `30-check-prognostic-confound-pd.R` a `--params` switch so that the placebo-response contrast can be recomputed under the published SDs.
- Give `37-ladder-tables-figures.R --scope all` a published-set input path if paper 15 is to carry ladder-style figures. Its trend figure (`02-ladder-fig4-trend.png`) was built from historical data and is not used here.

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A Absorption of a constant shift

Let X be the fixed-effects design matrix of any analysis here. It contains the biomarker main effect B_i^c. The constant shift adds $b_{TV}\sigma_{TV}z_i$ to every outcome of participant i. Since z_i is an affine function of B_i^c, the shift vector lies in the column space of X. For fixed variance parameters the generalized least-squares estimate then changes only in the coefficients of the intercept and B_i^c. The residuals are unchanged. REML and ML estimates of the variance parameters depend on the data only through those residuals, so they are unchanged too. AIC selection compares fits whose likelihoods are unchanged, so it selects the same half-life. CR2 standard errors depend on the residuals and the design only. The interaction estimate, its standard error and its p-value are therefore identical, replicate by replicate. The stored data confirm it (Section 3.2).

B The joint feasibility condition

Order the variables as biomarker, then everything else, and let M_0 be the correlation matrix of everything but the biomarker, including the baseline level, which is uncorrelated with the response components. By the Schur complement the joint matrix is positive definite if and

only if M_0 is and the biomarker row r satisfies $r^\top M_0^{-1} r < 1$. With c_{bm} alone, $r = c_{bm}\tilde{g}$, where $\tilde{g}$ is the correlation profile padded with zeros, and the bound is $c^*_{bm} = (\tilde{g}^\top M_0^{-1} \tilde{g})^{-1/2}$ (2).

With prognostic correlations the row is a sum of fixed patterns scaled by the correlations, $r = c_{bm}\tilde{g} + c_{bm,PB}\, p + c_{bm,TV}\, q + c_{bm,BL}\, b$. Here p and q are one on every visit of the placebo-response and natural-history components, and b is one on the baseline level. The condition becomes $c^\top A c < 1$, with $c = (c_{bm}, c_{bm,PB}, c_{bm,TV}, c_{bm,BL})$ and $A_{jk} = x_j^\top M_0^{-1} x_k$ for $x \in \{\tilde{g}, p, q, b\}$. The feasible region is the interior of an ellipsoid. Fixing all correlations but one makes $c^\top A c = 1$ a quadratic in that correlation, whose positive root is its bound. A negative discriminant means the fixed correlations are already infeasible. Because b is an eigenvector of M_0 with unit eigenvalue, $c_{bm,BL}$ enters only through its square. Its bound given c_{bm} is $\sqrt{1 - c_{bm}^2 A_{\tilde{g}\tilde{g}}}$. The p and q patterns are structurally identical, so $c_{bm,PB}$ and $c_{bm,TV}$ have the same bounds. A design's bound is the minimum over its paths. Table 3 gives the values for the Hybrid design. The off-diagonal entries of A are negative at the extended DGP's values, so the correlations are mildly complementary. The DGP sets the biomarker's marginal correlations with the other components to zero. The negative entries of A reflect partial association: given its correlation with the drug response, the biomarker's partial correlation with the other components is not zero. An explicit correlation of the same sign therefore uses less of the feasible region than it would if the components were independent.

C The shared biomarker effect

Let s_0, s_{off} and s_{on} be the true slopes of Y on B at baseline, at post-baseline off-drug visits and at on-drug visits. A model with baseline as an off-drug row estimates one β_B for s_0 and s_{off}.

The following account is heuristic, not a derivation. Consider a prognostic biomarker with no interaction, whose post-baseline slope is constant: $s_0 = 0 \neq s_{\text{off}} = s_{\text{on}} = s_{\text{prog}}$.

- **The bias.** The fitted $\hat{\beta}_B$ lies between the baseline and off-drug slopes, $\hat{\beta}_B \approx f\, s_{\text{prog}}$ for some $0 < f < 1$ that depends on the weight the baseline row receives. The interaction absorbs the remainder, $\hat{\beta}_{bm:D} \approx (1 - f)\, s_{\text{prog}} \neq 0$.
- **The correction.** Adding $B_i \cdot \mathbf{1}\{t = 0\}$ frees s_0 and removes this source of bias. The constraint is analogous to the choice, in longitudinal analysis, between treating baseline as a covariate and as a response (15).

Two further mechanisms operate in the scenarios of Section 3.6.

- **The slope grows over time.** When the post-baseline slope grows, the bias depends on how drug status is confounded with time. Freeing s_0 is then not sufficient in general. The rejection rate of 0.066 for that analysis under the growing pathway is consistent with this. Section 3.6.1 shows that the same growing shift biases the interaction with or without a baseline row, in opposite directions.

- **Expectancy.** When the biomarker is associated with the placebo response, the slope also scales with the expectancy η_t, which is higher at the open-label (on-drug) visits. The expectancy term $B_i \cdot \eta_t$, with $\eta_0 = 0$, addresses both this mechanism and the baseline constraint. Freeing s_0 alone does not. An expectancy-scaled slope still enters the interaction through the open-label visits.

Neither correction is exact in the Hybrid design. The expectancy term removes a constant post-baseline slope from the interaction only when the blinded on-drug and off-drug visits carry equal total weight. In the Hybrid design they do not, so a small bias remains. The corresponding derivations are in the project's compendium of mathematical results (`analysis/report/compendium-mathematics.tex`). The null rejection rates of Section 3.6 across the four prognostic cells were 0.040 to 0.055 for the expectancy-term model and 0.051 to 0.066 for the free-baseline model. The residual bias of the expectancy-term model is therefore within Monte Carlo error at the prognostic correlations simulated. That of the free-baseline model is not, under the growing pathway. Under a prognostic association with baseline severity alone the binary-indicator analysis rejected 0.064, and the two corrections rejected 0.055 and 0.054.

D First run of the second stress test

The first run of the stress test of Section 3.6 used the compounding carryover rule on the mean and fixed path allocation (18, 18, 17, 17). Its table is given here for comparison with Table 10. The two runs differ only in the carryover rule.

E Draft consistency checks

The prose of this draft makes 40 qualitative statements that are checked against the stored data when it renders. All of them hold.

Table 17: Second stress test, first run (compounding carryover rule, fixed allocation): null rejection rates under a prognostic biomarker, 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Rates above 0.075 in bold.

Analysis	Natural history, constant	Natural history, growing	Placebo response	Baseline severity
Binary-indicator analysis (random intercept, model-based)	**0.107**	0.053	**0.159**	0.075
Random intercept + CAR(1), CR2	**0.099**	0.049	**0.100**	0.050
The same, half-life chosen by AIC	**0.115**	0.058	**0.112**	0.055
Unstructured covariance, Kenward-Roger	**0.416**	**0.195**	**0.222**	0.051
The same, half-life chosen by AIC	**0.509**	**0.245**	**0.256**	0.064
Random intercept + CAR(1) with expectancy terms, CR2	0.047	0.046	0.040	0.040
The same, with phase-specific residual variances, CR2	0.055	0.044	0.042	0.042
The same, with phase-specific residual variances, model-based	0.062	0.043	0.045	0.046
Expectancy terms, half-life chosen by AIC, CR2	0.059	0.052	0.047	0.045
Random intercept + CAR(1) with free baseline slope, CR2	0.049	0.063	0.049	0.050